

String Objects

Tasks

In this task you will be exploring the Javascript built-in *String Object*, along with some of its common methods for doing useful work.

To complete this task, you are asked to use the provided simple form that obtains from the user three inputs (an input string, a string to search for, and the number of characters setting the size of a substring). Your code should then, based on this input, provide the output from various *string object* operations. The actual layout/design of the web page is left up to you, but you can use the sample provided in the supplied code, which displays as shown below:

The screenshot shows a web browser window with the title 'Ice Cream T-Shirt Printing'. The address bar shows 'localhost:3000'. The page has a light blue header with the text 'World of Ice Cream' and 'Putting text on our Ice Cream T-Shirts'. Below the header is a section titled 'Ice Cream T-Shirt Printing - Text Processing'. The instructions say: 'Please enter into the text-box below, what you would like displayed on your T-Shirt. Then click the `Process Text` button to show some interesting points about your text.' There is an 'Example textarea:' with a text input field. Below that is a 'Keyword to search for:' with a dropdown menu showing 'Ice Cream'. Then 'Number of characters in sub string (between 10 and 50):' with a text input field showing '10'. There are two buttons: 'Process Text' (blue) and 'Reset Text' (grey). Below the buttons is a table with two columns: 'String operation...' and 'Result obtained...'. The table has seven rows, all of which are currently blank. At the bottom of the page is a footer that says '© 2024 Ice Cream made chilled!'.

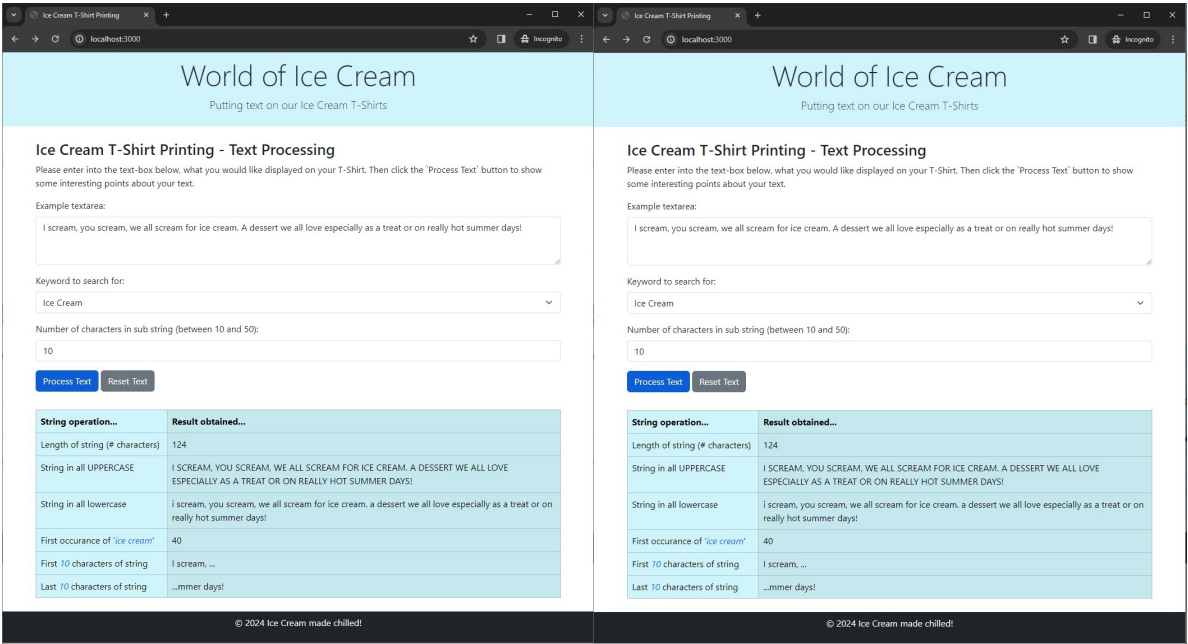
String operation...	Result obtained...
Length of string (# characters)	
String in all UPPERCASE	
String in all lowercase	
First occurrence of '??'	
First _ characters of string	
Last _ characters of string	

Input Form and Blank Results

The image below shows the output given with an input string:

```
`I scream, you scream, we all scream for ice cream. A dessert we all love especially as a treat or on really hot summer days!`
```

Followed by the search string obtained from a selection input, and finally a number input form a range between 10 and 50 to be used as the size of the substrings (one from the start of the input string, the other from the end of the input string). In the *left* example, the search string is `Ice Cream` and the substring size is `10`, while the *right* example has a search string of `Love` and a substring size of `32`.



Data input and results

If the search string provided does not exist in the input string then a value of `-1` should be displayed, as shown below:

World of Ice Cream

Putting text on our Ice Cream T-Shirts

Ice Cream T-Shirt Printing - Text Processing

Please enter into the text-box below, what you would like displayed on your T-Shirt. Then click the 'Process Text' button to show some interesting points about your text.

Example textarea:

I scream, you scream, we all scream for ice cream. A dessert we all love especially as a treat or on really hot summer days!

Keyword to search for:

Fun

Number of characters in sub string (between 10 and 50):

50

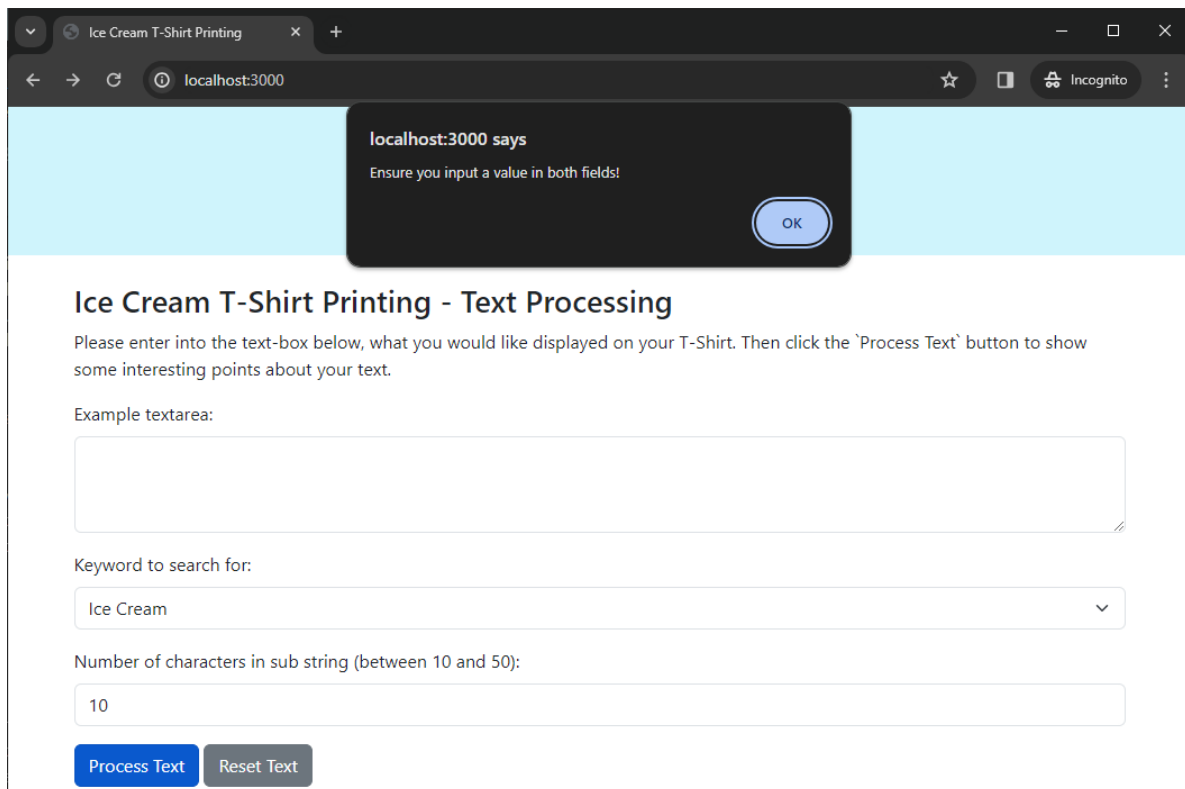
[Process Text](#) [Reset Text](#)

String operation...	Result obtained...
Length of string (# characters)	124
String in all UPPERCASE	I SCREAM, YOU SCREAM, WE ALL SCREAM FOR ICE CREAM. A DESSERT WE ALL LOVE ESPECIALLY AS A TREAT OR ON REALLY HOT SUMMER DAYS!
String in all lowercase	i scream, you scream, we all scream for ice cream. a dessert we all love especially as a treat or on really hot summer days!
First occurrence of 'fun'	-1
First 50 characters of string	I scream, you scream, we all scream for ice cream....
Last 50 characters of string	...specially as a treat or on really hot summer days!

© 2024 Ice Cream made chilled!

Data input and no search string found

Additionally, if **no** input string or search string is provided, an **alert** message should be displayed, as shown in the screenshot below:



localhost:3000 says
Ensure you input a value in both fields!

OK

Ice Cream T-Shirt Printing - Text Processing

Please enter into the text-box below, what you would like displayed on your T-Shirt. Then click the 'Process Text' button to show some interesting points about your text.

Example textarea:

Keyword to search for:

Ice Cream

Number of characters in sub string (between 10 and 50):

10

Process Text Reset Text

Data input and no search string found

Finally, if the `Reset Text` button is pressed, the input and result fields should all be cleared to match the initial state of the page.

Steps

The provided web page has the respective **input** and **output** fields individually identified using the HTML `id` attribute. These can be used to obtain the values for processing, as well as for updating the table.

In example provided, the results fields start off as being empty and then when the user clicks the `Process Text` button, the respective fields are updated with the output from the various *string methods*.

Each of the input string variables will be stored as a javascript `String` inbuilt object that comes with a number of helpful/useful methods (functions) to perform work.

The specific *string* operations to perform on the *input string* include:

1. Display the total **number of characters** from the provided *input string*.
2. Display all characters of the *input string* in **UPPERCASE**.
3. Display all characters of the *input string* in **lowercase**.
4. Display the **beginning position** of the *input substring* found within the *input string* (should be -1 if not found)

NOTE: Convert the search string into a *lowercase* and then search in the *lowercase input string* to be case **insensitive** (There are other ways to do this and you are welcome to implement these if you like!)

5. Display the **first *n* characters** of the *input string*, where *n* is provided by the user.
6. Display the **last *n* characters** of the *input string*, where *n* is provided by the user.

Hints

As done in the tasks from Week 06, create a javascript file that holds **two** functions that are called on the event one of the buttons is clicked (e.g., one for `processText()` and the other for `clearText()`). You should be able to use the unique IDs to the respective input and output fields within the *Input Data* form and *Processed Results* section, to extract/update these fields from the javascript code.

The `alert` error message is displayed when either the *input string* **OR** the *search string* is empty, e.g.:

```
if( inputString == "" || searchString == "" ) {
    ... alert error message ...
} else {
    ... process the form input and
    ... display the results
}
```

Furthermore, review the various methods/functions available to work with *JavaScript String Objects*:

- https://www.w3schools.com/js/js_string_methods.asp
- https://www.w3schools.com/jsref/jsref_obj_string.asp

References for JS pop-up boxes, including `alert()`:

- https://www.w3schools.com/js/js_popup.asp

What will you submit?

You should submit:

- Screen-shot of the web page with valid input, search string and substring length.
- Screen-shot of the web page with valid input, a different search string and different substring length.

SIT120 Introduction to Responsive Web Apps

- Screen-shot of the web page with valid input and non-existent substring.
- Screen-shot of the web page missing the input and showing the `alert` error message.
- Your HTML file.
- Your JavaScript file.